

HIBERNATION



H DURING WINTER, as temperatures drop and food becomes scarce, some animals hibernate to survive the harsh conditions. Hibernation is a resting state in which the animal's body temperature falls to just above that of its surroundings, and its metabolic rate (the rate at which it consumes energy) drops dramatically. The animal resumes its active lifestyle in the spring. Hibernation is triggered by shortening day length, a fall in temperature, or by the animal's internal biological clock. Some other animals rest, or remain dormant, to withstand adverse conditions.

Rodents

Rodents, such as dormice and woodchucks, form the largest group of hibernating mammals. Many smaller rodents living in the northern hemisphere hibernate in the winter months when the plant material and small animals they feed on are in short supply. Some construct nests in tree hollows or underground in which they curl up, to minimize heat loss, and go into a deep sleep. Many species wake periodically either to eat, drink, or urinate, to get rid of accumulated waste.



Nest of straw and grass

Hibernating dormouse

Dormouse curled up in its nest

Hibernation

Rodents, bats, and insectivores are all mammals that hibernate. Their small size allows them to cool down and warm up quickly. Some animals eat more in early winter to build up fat stores to use while hibernating; others wake every few weeks to feed on food in their nest. A squirrel shows changes typical of a hibernating mammal; its metabolic rate drops to 1 per cent of normal, and its body temperature falls from 37°C (99°F) to 4°C (39°F).

When it emerges its body weight will have fallen by 40 per cent.

Natterer's bat



Bats

Many temperate species of bat hibernate when their insect food disappears. They often hibernate in large numbers and cluster together to conserve warmth. The site where bats hibernate is called a hibernaculum; it may be a cave, mine, tree hollow, or a deserted building.



Hummingbird in nest

Birds

Most birds migrate to avoid cold winters; a few such as the North American poorwill enter a state of torpor and hibernate. Many hummingbirds show a form of daily hibernation. At night, their body temperature falls, enabling them to survive cooler conditions without consuming much energy.



European badger

Badgers live in forests where they dig extensive burrows called setts. In winter, badgers rarely leave their sett. They curl up in nesting material and go into a dormant state, living off fat reserves accumulated in summer and autumn. Dormancy can last for seven months in Siberia.

Dormancy

Some large mammals, such as bears and badgers, that live in northern parts of North America and Europe, go into a resting state, called dormancy, during winter. Dormancy differs from hibernation; the animal's body temperature does not drop significantly, and it can wake up quickly if danger threatens. However, this small fall in body temperature, combined with a lack of activity, produces significant energy savings for the animal.

Bear feeds on berries to build up fat reserves.



Brown bear

The brown bear lives in Asia, North America, and Europe. In summer it builds up fat on which it lives in winter. In autumn, the bear excavates a den, lines it with vegetation, and goes into its winter "sleep". Its body temperature falls by 5°C (9°F), and its metabolic rate drops by 50 per cent. The bear emerges in spring weighing half what it did in the autumn.



Cabbage white butterfly pupa

Diapause

Diapause is the insects' equivalent of hibernation. Some insects enter diapause to survive adverse conditions, such as cold or lack of food. During diapause, growth and development are suspended, usually at the egg or pupa stage of the life cycle. For example, if the cabbage white butterfly lays her eggs in late summer, the pupal stage goes into diapause over winter, resuming development in spring.

Aestivation

Aestivation is a state of dormancy, shown by animals such as African lungfish, during hot, dry summers. Lungfish live in places that flood in the wet season and bake in the dry season. As river levels fall, the lungfish digs a burrow in the mud, ending in a chamber. The fish curls up in the chamber, secretes a protective mucus bag around itself, and remains there for up to six months until the rains come.



African lungfish

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